

Growing Brassicas in the Dragoon Mountains

Guide J · Cool-Season Track

Broccoli · Cauliflower · Cabbage · Kale · Brussels Sprouts · Kohlrabi

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RELATED GUIDES

This guide opens the **Cool-Season Track** of the Dragoon Mountains Growing Library — the half of the year when peppers and tomatoes are off the field.

- **Guide A — Growing Peppers in the Dragoon Mountains** · the warm-season companion
- **Guide B — Bato Bucket Hydroponic Setup** · the system used in the hydroponic section below
- **Guide D — Hydroponic Primer** · water, nutrient, and system basics referenced throughout
- **Guide F — DIY Hydroponic Nutrients** · the locally-sourced three-part formula adapted here for brassicas
- **Guide G — pH Up & pH Down Replacements** · for tuning the alkaline well water that's typical in Cochise County

How to use this guide

Brassicas are the **cool-season anchor crop** for the Dragoons. Pepper season runs roughly April through October; brassicas fill the remaining months. The two crops never compete for bed space, and they reward very different management — so this guide is built to be used alongside Guide A, not in place of it.

The guide is in three parts:

1. **Common ground** — climate window, variety selection, and the brassica-specific basics that apply whether you grow in soil or in solution.
2. **Soil & containers** — bed prep, transplant timing, pest pressure, and crop-by-crop harvest notes.
3. **Hydroponics** — bato bucket and NFT recipes for the same crops, with EC targets and nutrient adjustments for cool-temperature heads.

A short troubleshooting section closes the guide.

LOCAL TIP

Every brassica in this guide tolerates light frost and most actually improve with it — broccoli, kale, and brussels sprouts in particular sweeten after the first few hits below 32°F. Plan your harvest *after* the first frost, not before.

Part 1 — Common ground

1. The Dragoons cool-season window

The Dragoon Mountains sit at roughly 4,700 ft in USDA zone 8a. Daytime highs from late September through April are usually in the 50s–70s, with overnight lows from the high 30s into the low 20s during the deepest weeks of January. Hard freezes below 25°F happen but are short. This is, by national standards, an extremely generous brassica window: you can plant in late August for fall heads, again in February for spring heads, and overwinter the hardier crops (kale, brussels sprouts, savoy cabbage) through the whole period.

The trade-off is **soil temperature and water**. The ground stays warm well into October, which delays root establishment if you transplant too early; and winter watering still matters because rainfall is minimal and brassicas in the Dragoons rarely get more than a half-inch of natural precipitation per month from October through April.

2. Variety reference

Pick varieties bred for short, cool windows. Long-season European cultivars often bolt or fail to form heads here because our cool window is mild rather than long.

Crop	Recommended varieties · profile	Dragoon notes
Broccoli	Calabrese · open-pollinated · 60–65 days · loose mid-size heads with abundant side shoots · classic spring crop. De Cicco · 50–60 days · small main head, heavy side-shoot producer · best choice for a long picking window. Belstar F1 · 65–75 days · large tight heads, holds in heat better than most.	Calabrese and De Cicco both keep producing side shoots for 4–6 weeks after the main cut — favored for fall plantings where you want a long harvest before deep frost. Belstar’s heat tolerance is useful for late-summer sowings.

Crop	Recommended varieties · profile	Dragoon notes
Cauliflower	Snowball Y · open-pollinated · 60-70 days · classic white head, needs blanching (tie outer leaves over the head). Cheddar F1 · 60-65 days · orange self-blanching, no leaf-tying needed · sweet flavor. Graffiti F1 · 75-85 days · purple, holds color when cooked briefly.	Cauliflower is fussier than broccoli — needs steady moisture and a narrow temperature band (55-65°F) during head formation. The Dragoons' warm fall makes early sowings risky; aim for transplants going out late September. Self-blanching varieties save 10 minutes per head per week of fussing with twine.
Cabbage	Early Jersey Wakefield · open-pollinated · 60-65 days · conical heads, very early. Late Flat Dutch · 100 days · large 8-10 lb flat heads, storage cabbage. Savoy Ace F1 · 80 days · crinkled leaves, most cold-hardy.	Wakefield is the spring workhorse here — small enough to size up before the May heat. Savoy Ace overwinters dependably; the heads slow but don't damage through low-20s nights. Storage cabbages need the full 100-day window — plant by September 1 for Christmas harvest.
Kale	Lacinato (dinosaur) · 50-65 days · narrow dark blue-green strap leaves · holds through hard frost. Red Russian · 50-60 days · purple-veined oak-leaf shape · most cold-hardy and most tender. Winterbor F1 · 60 days · classic curly green, very productive.	Kale is the easiest brassica in the Dragoons — direct-sow in September, harvest continuously through April. Frost concentrates the sugars; midwinter kale tastes sweeter than fall kale. Red Russian self-sows freely if you let a few plants bolt in May.

Crop	Recommended varieties · profile	Dragoon notes
Brussels sprouts	Long Island Improved · open-pollinated · 90 days · classic open-pollinated standard. Diablo F1 · 110 days · tight uniform sprouts, holds long on stalk. Falstaff F1 · 100 days · purple, decorative and flavorful.	Brussels sprouts need the longest run of any brassica — direct-sow or transplant in early August, harvest October through January after multiple frosts. Top the plant (remove growing tip) in late October to push energy into the sprouts. The hardest part is keeping aphids off in November.
Kohlrabi	Early White Vienna · 55 days · pale green, the standard. Purple Vienna · 60 days · purple skin, white flesh. Kossak F1 · 80 days · stores in the ground without going woody up to grapefruit size.	Underrated for the Dragoons — fast, takes light frost, edible at any size from golf ball to softball. Direct-sow in succession every three weeks from late August through early November for a winter-long supply.

Brassicas hate heat at heading time. The single most common failure mode in the Dragoons is transplanting cauliflower or broccoli too early in the fall — warm soil pushes leaf growth, then a hot Indian-summer week in mid-October sends the plants into “buttoning,” where they form a tiny premature head and stop. Wait for the 10-day forecast to drop highs below 80°F before you set out transplants.

3. Timing windows for the Dragoons

Sow indoors	Transplant out	Direct sow	Harvest window	Best for
Mid-July to early August	Late August to mid-September	Aug 25 - Sep 25	October - January	Broccoli, cauliflower (early), kohlrabi, cabbage (early), kale

Sow indoors	Transplant out	Direct sow	Harvest window	Best for
—	—	Sep 1 - Oct 15	November - April	Kale, kohlrabi succession, mâche, mustards
Late November	Mid-January under row cover	—	March - May	Spring broccoli, spring cauliflower, early cabbage
Early February	Late February to mid-March	Late Feb - early April	April - June	Spring cabbage, kohlrabi, broccoli raab

LOCAL TIP

The Dragoons sit at the edge of where you can get *two* full brassica crops per year — one fall/winter, one spring. If you only want one, fall is more reliable. Spring brassicas race against May heat and the wind that picks up in April; some varieties bolt before they head.

Part 2 — Soil & containers

4. Bed preparation

Brassicas are heavy feeders. They want a loose, fertile bed with steady moisture and a near-neutral pH. The native Dragoons soil is alkaline (typically pH 7.6–8.1), low in organic matter, and drains too fast on slope or too slowly in caliche pockets — none of which suits a broccoli plant well. Bed prep is where most of the work happens.

Soil chemistry targets

Parameter	Target for brassicas	Typical Dragoons baseline	Adjustment
pH	6.5 – 7.0	7.6 – 8.1	Acidify with elemental sulfur (1 lb per 100 sq ft drops pH ~0.5 point over a season) or annual compost amendments
Organic matter	5 – 8%	1 – 2%	2–3 inches of finished compost worked into the top 8 inches every fall
Nitrogen	100–150 lb/acre equivalent	low	Aged manure or feather meal at transplant + a side-dress at heading
Calcium	High; cauliflower is especially needy	naturally high in Dragoons soil	Usually no addition needed — caliche supplies it
Boron	1–2 ppm	typically low	A pinch (1/4 tsp) of borax dissolved in 1 gal water, applied per 50 sq ft, prevents hollow stem
Soil temp at transplant	55–70°F	75°F+ in early fall	Wait, or shade-cool the bed for 5 days before planting

Practical bed routine

1. **Late July:** Remove pepper plants from the summer bed or pull the cover crop. Pull weeds while they're young.
2. **Late July / early August:** Spread 2–3 inches of finished compost over the bed. If the soil tested alkaline last year, broadcast 1 lb elemental sulfur per 100 sq ft now.
3. **Early August:** Fork or broadfork to 8 inches. Avoid rototilling — you destroy soil structure that brassicas reward.
4. **Mid-August:** Water the bed deeply twice in the week before planting. This wakes the soil biology and cools the surface.
5. **Late August:** Test soil temperature in the morning with a probe. When it's below 75°F at 4-inch depth, you can transplant.

5. Direct sow vs transplant

Decision	When to direct-sow	When to transplant
Kale, kohlrabi, mustards, broccoli raab	Almost always — they germinate fast and don't mind being thinned	If you need precise spacing in a tight bed
Broccoli, cauliflower, cabbage	If you've got time and the warm-soil window is still 6+ weeks before frost	The default — buys you 4–6 weeks of growth and avoids the seed-eating birds (white-crowned sparrow, lark sparrow) that hammer bare beds in late August
Brussels sprouts	Never — they need every week of the season	Always — start indoors in early July if you can

For transplants, sow into 4-inch pots in a sheltered shady spot (or indoor under lights at 65–70°F). Pot up once if they outgrow before transplant date. Harden off for 5–7 days before setting out.

6. Spacing & companion planting

Crop	In-row spacing	Between rows	Notes
Broccoli, cauliflower	18 in	24–30 in	Tight spacing slightly reduces head size but boosts overall yield per square foot
Cabbage (small/early)	12–15 in	18 in	Wakefield-type can go 12 in

Crop	In-row spacing	Between rows	Notes
Cabbage (large/storage)	18-24 in	24 in	Late Flat Dutch needs 24 in
Brussels sprouts	24 in	30 in	They grow tall and topple — stake or plant in a sheltered row
Kale	12 in	18 in	Tight spacing keeps plants tender; wider gives bigger leaves
Kohlrabi	6 in	12 in	Closer than you'd think — they don't compete much

Good companions in adjacent rows: alliums (onion, garlic) repel cabbage moth; dill and cilantro flowers feed the parasitic wasps that kill brassica pests; nasturtiums act as an aphid trap. Avoid following or preceding brassicas with other brassicas in the same bed (4-year rotation is the goal); avoid pairing with strawberries.

7. Pest management

The Dragoons have a moderate brassica pest load — fewer pests than coastal California, more than the Rockies. Three matter most.

Cabbage white butterfly (*Pieris rapae*) and cabbage looper

Small white butterflies appear in August, lay eggs on leaf undersides; the green caterpillars eat fast and leave dark frass. Control:

- **Floating row cover** at transplant, until the plants are big enough that perforations don't matter (about 4 weeks). Cover edges with soil; the butterflies will lay through any gap.
- **Bt (*Bacillus thuringiensis*) spray** weekly if row cover comes off. Spray at dusk; Bt breaks down in UV.
- **Hand picking** is realistic for small plantings — 10 minutes a week per 50 sq ft.

Harlequin bug (*Murgantia histrionica*)

Distinctive orange-and-black shield bugs, the dominant fall pest in the Dragoons. They suck sap and cause yellow-and-white stippling on leaves; heavy populations stunt or kill young plants. Control is harder than for caterpillars because Bt doesn't work on them.

- **Hand picking morning and evening** — they drop when disturbed; hold a jar of soapy water under the plant and tap.

- **Trap crops** — leave a row of mustard or radish flowering nearby; harlequins prefer it and you can vacuum the flowers daily.
- **Neem or pyrethrin** for severe infestations only; both kill beneficial insects too.

Aphids (cabbage aphid, green peach aphid)

Late fall and overwintered plants get hammered by gray-green aphids that cluster in growing tips. Cold weather actually favors them — they outpace their predators in November.

- **Hard water spray** every 3 days knocks populations down. Concentrate on growing tips.
- **Insecticidal soap** for resistant clusters; reapply weekly.
- **Tolerate moderate populations** on outer leaves — you'll strip those at harvest anyway.

WARNING

Do **not** spray pyrethrin or neem when bees are active and avoid spraying anything blooming. Brassicas in flower (bolted, or seed crops) are major late-fall and early-spring pollinator forage in the Dragoons. If you must spray, do it at dusk and only on plants not in flower.

8. Per-crop harvest notes

Broccoli

Cut the main head when the buds are tight, dense, and dark green, before any yellow shows. The stem should be tender enough to push a thumbnail in. Cut 5–6 inches below the head with a sharp knife at a 45° angle (so water doesn't pool on the cut surface). Leave the plant — side shoots form at every leaf axil for 4–6 weeks after the main cut, especially on Calabrese and De Cicco.

Cauliflower

When the head is the size of a tennis ball, gather the outer leaves over it and tie loosely with twine or a rubber band — this is “blanching” and keeps the head white. Self-blanching varieties (Cheddar, Snow Crown) wrap themselves. Check daily once heads start to size up; cauliflower goes from perfect to ricey-loose in 3 days at the wrong temperature. Cut the head with a long handle of leaves attached for stem moisture.

Cabbage

Heads are mature when the top feels firm to a fingertip-press and the outer wrapper leaves start to peel back slightly. Cut at the base with a knife, leaving the root and lowest 2 leaves in the ground — small secondary heads will form at the cut and can

be harvested in 4–6 weeks. Heads will hold in the ground for weeks in cool weather; they're done when they split (a sign of water uptake after maturity).

Kale

Harvest the outer (largest, lowest) leaves first, leaving the central growing tip and at least 6 leaves to keep producing. Pick continuously from October through April. After hard frost, the leaves taste markedly sweeter; pick frequently to keep new tender leaves coming.

Brussels sprouts

Sprouts mature from the bottom of the stalk upward. Pick when they're firm and 1–1.5 inches across, snapping each off by hand or with a small knife. In late October, top the plant (cut off the growing tip and 1–2 small upper leaves) — this stops new sprouts forming and concentrates energy into the existing ones, producing more uniform size. Harvest can continue into January.

Kohlrabi

Pull when the bulb (actually a swollen stem) is 2–4 inches across — they go woody above tennis-ball size, except for Kossak which holds. Slice off roots and leaves; the bulb stores 2 months in the fridge. Don't compost the leaves — they're more nutritious than the bulb and sauté like collards.

Part 3 — Hydroponics

Brassicas are well-suited to hydroponics in the Dragoons, particularly in the cool months when an outdoor or covered-patio bato bucket setup runs at near-ideal head-forming temperatures. The advantage over soil here is precision: you can hit the calcium, magnesium, and boron numbers exactly, and brassicas grown with those nutrients dialed in form tighter, denser heads.

9. System selection

System	Suits	Caveats
Bato (Dutch) bucket	Broccoli, cauliflower, cabbage (small varieties), brussels sprouts	Best all-rounder for the bigger plants. See Guide B for setup.
NFT channel	Kale, kohlrabi, lettuce-bed crops	Don't try heading brassicas in NFT — the channels are too small for root mass.
Deep water culture (DWC)	Kale, mustards, broccoli raab	Cheap and effective for leaf crops; not recommended for heading types.
Drip-to-drain in coco	All brassicas	A reasonable middle ground — easier than bato but slower-growing.

For broccoli and cauliflower, **bato buckets are the recommended system** in the Dragoons. The volume per plant (around 3 gallons of media) supports the root mass; the drip-and-drain cycle keeps oxygen in the root zone; and the bucket spacing matches the 18-inch plant spacing nicely.

10. Nutrient targets

Brassicas want **less nitrogen and more potassium and calcium** than peppers or tomatoes. They also have a higher boron and sulfur demand than most crops. Start with the Guide F three-part recipe and adjust as below.

Stage	EC target	pH	N-P-K	Notes
Transplant to 4 true leaves	1.2–1.4 mS	6.0–6.3	Higher N for leaf bulk	Standard Guide F mix at 75% strength
Vegetative (rosette)	1.6–1.8 mS	6.0–6.3	Balanced	Full Guide F mix

Stage	EC target	pH	N-P-K	Notes
Pre-heading (first sign of curd)	1.8-2.0 mS	6.0-6.5	Reduce N, add K	Cut Part A by 20%, add 0.3 g/L potassium sulfate (or wood-ash potash from Guide F)
Heading	1.8-2.0 mS	6.0-6.5	Same	Maintain — high N at this stage causes loose heads

Boron and calcium together prevent the two biggest brassica disorders.

Hollow stem (broccoli, cauliflower) and tipburn (cabbage, brussels sprouts) are both deficiency-driven and both common in hydroponics where the medium doesn't supply trace minerals. Add a foliar spray every 10 days during heading: 0.5 g borax + 1 g calcium chloride per liter of water, spray on undersides of leaves at dusk.

11. Temperature management

This is the hardest part. Brassicas form heads best at 55-65°F air temperature, 60-68°F root-zone temperature. In the Dragoons:

- **Fall (Oct-Dec):** Outdoor or covered-patio bato setups run at near-perfect temperatures during the day; nights drop to the high 30s which slows growth but doesn't damage the plants. Cover with frost cloth on nights below 25°F.
- **Winter (Jan-Feb):** Reservoir temperature is the issue — solution below 50°F slows nutrient uptake and invites root pathogens. Either insulate the reservoir or move it indoors. A 100W aquarium heater set to 60°F handles a 25-gallon reservoir.
- **Spring (Mar-May):** The opposite problem — afternoon highs hit 80°F by mid-April and stress the plants. Shade cloth (30%) over the system extends the season by 2-3 weeks.

LOCAL TIP

The same bato bucket array used for summer peppers (Guide C) can run cool-season brassicas with no rebuilding. Drain the reservoir at season turnover, scrub with weak hydrogen peroxide, refill, and switch to the brassica nutrient profile. The hardware doesn't care which crop it's holding.

12. Crop-specific hydroponic notes

Broccoli (bato bucket)

Calabrese and De Cicco perform exceptionally well — slightly smaller heads than soil-grown but denser and earlier. Expect 50–55 days from transplant to first cut in bato. Side shoots are even more prolific than in soil because nutrient delivery doesn't quit. Plan on 8–10 weeks of total harvest per plant.

Cauliflower (bato bucket)

The most demanding hydroponic brassica. Curd quality is highly sensitive to nutrient swings — keep the reservoir EC and pH stable to within 0.1 unit of target during head formation. Self-blanching varieties (Cheddar, Snow Crown) are strongly recommended.

Cabbage (bato bucket, small varieties only)

Wakefield and other compact varieties size up well; storage cabbages get root-bound. Heads are ready 10–14 days earlier than soil-grown, but storage life is shorter — eat them fresh or process within 2 weeks.

Kale (NFT or DWC)

The easiest crop to run hydroponically. Continuous harvest from week 4 onward; replace plants every 12–14 weeks as they exhaust. Red Russian and Lacinato both thrive; curly kales are slightly slower.

Brussels sprouts (bato bucket)

Tall plants need bucket stability — fill the bottom of the bucket with extra perlite or pea gravel for ballast and stake the plant to the bucket support. Otherwise grows similarly to soil but produces sprouts about 2 weeks earlier.

Kohlrabi (any system)

Extremely tolerant — runs in bato, NFT, DWC, or coco-drip with similar success. Quick crop (50 days), can be slotted between heading-brassica successions.

Troubleshooting

Symptom	Likely cause	Fix
Broccoli plant grows large, forms tiny premature head and stops ("buttoning")	Transplanted into warm soil, then hit by a hot week during head initiation	Push transplant date later; provide shade if heat is forecast during early heading
Cauliflower head is loose, ricey, or yellow	Temperature swings during head formation; insufficient blanching; over-fertilization with N	Use self-blanching varieties; tie outer leaves promptly; cut N by 20% at heading
Brown or hollow center in broccoli or cauliflower stem	Boron deficiency	Foliar spray 0.5 g borax/L every 10 days during heading; add boron to nutrient solution at 0.3 ppm
Brown leaf edges on cabbage or brussels sprouts ("tipburn")	Calcium delivery interrupted (uneven watering, or low transpiration during overcast cool weather)	Steady irrigation; foliar calcium chloride spray at 1 g/L weekly
Kale leaves turn purple-tinged and slow growth	Phosphorus deficiency triggered by cold root zone	Warm reservoir to 60°F; in soil, side-dress with bone meal or fish bone meal
Plants form leaves but no head all winter	Variety not suited to short cool windows (too long a maturity for our climate)	Switch varieties — pick under 70 days to maturity
Cabbage head splits while still on plant	Sudden water uptake after maturity	Harvest immediately; for future plantings, cut back water as heads firm up
Aphid populations explode in late fall	Cold weather slowed predators faster than aphids	Hard-water spray every 3 days; trim heavily infested leaves and discard (don't compost)

Closing

Brassicas reward steady attention more than effort. Once the bed or the bucket is prepped and the transplants are in, the management is mostly observation — checking soil temperature, scouting for harlequin bugs, walking the row at dusk. A fall and winter of broccoli, kale, cabbage, and brussels sprouts more than fills the gap when the peppers come out, and the rotation does good things for the bed too — brassicas leave the soil structure improved and the nitrogen drawn down, which is exactly where you want it before next spring's peppers.

RELATED GUIDES

Cool-Season Track companions (planned):

- **Guide K — Lettuce & Salad Greens** · NFT and outdoor succession through the cool months
- **Guide L — Root Crops** · carrots, beets, turnips, and radishes in the Dragoon winter
- **Guide M — Alliums** · garlic, shallots, and overwintering onions
- **Guide N — Cover Crops** · cool-season legumes and cereal rye for the fallow peppers bed

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